

Supplementary Material

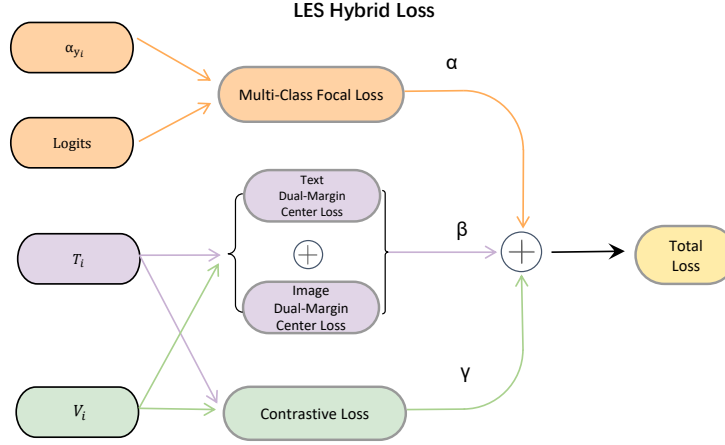


Figure 1: The detailed structure of LES Hybrid Loss.

Figure 1 above illustrates the detailed structure of the LES Hybrid Loss employed in our study. For a comprehensive explanation of this loss function, please refer to the main text.

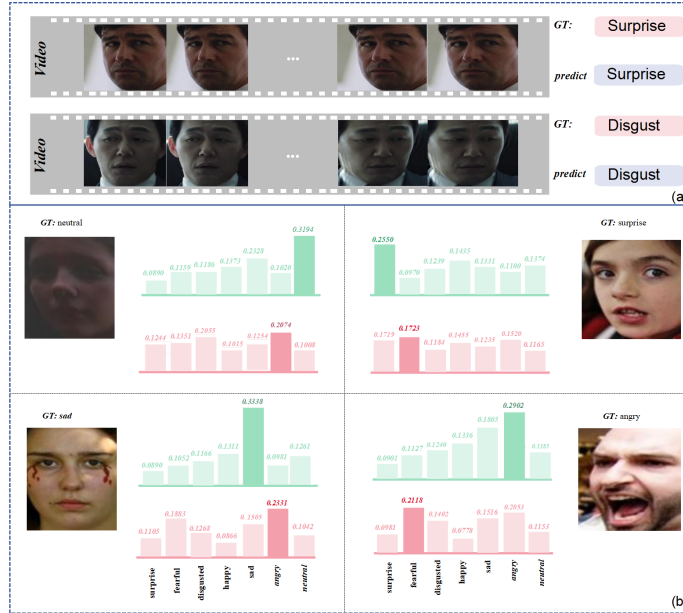


Figure 2: This figure presents a detailed analysis of our model's test performance on individual case examples. (a) illustrates LES-CLIP's predictions on dynamic FER samples from challenging or low-frequency emotion categories. (b) compares the fine-grained responses of LES-CLIP and CLIP on samples from static FER datasets.

Figure 2 demonstrates that LES-CLIP excels in dynamic FER by accurately distinguishing subtle emotional transitions that are often ambiguous even to human observers. In static FER tasks, LES-CLIP consistently outperforms the original CLIP, particularly on confusable and low-frequency categories, achieving more precise and balanced emotion classification. In the figure, red bars represent the predicted class

probabilities from CLIP. While green bars indicate the corresponding outputs from LES-CLIP, highlighting its enhanced emotional sensitivity.